



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Course: Fundamentals of Data Science

Course Code: DSA0404

Slot: A

Faculty: KRISHNARAJ M

Submitted By

Y Sravan Kumar(19247228)

Lab Programs

9.Real Estate Property

CODE

```
RealEstate.py - C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/RealEstate.py (3.12.3)
File Edit Format Run Options Window Help

import pandas as pd

# Input data
property_data = pd.DataFrame({
    'Property_ID': [101, 102, 103, 104, 105],
    'Location': ['Chennai', 'Bangalore', 'Chennai', 'Hyderabad', 'Bangalore'],
    'Bedrooms': [3, 5, 4, 6, 2],
    'Area_sqft': [1200, 2500, 1800, 3000, 1000],
    'Listing_Price': [6000000, 12000000, 8500000, 15000000, 5000000]
})

# 1. Average listing price in each location
avg_price = property_data.groupby('Location')['Listing_Price'].mean()

# 2. Number of properties with more than four bedrooms
count_properties = (property_data['Bedrooms'] > 4).sum()

# 3. Property with the largest area
largest_property = property_data.loc[property_data['Area_sqft'].idxmax()]

print("1. Average Listing Price by Location:")
print(avg_price)

print("\n2. Number of Properties with More Than Four Bedrooms:")
print(count_properties)

print("\n3. Property with the Largest Area:")
print(largest_property)
```

OUTPUT

```
IDLE Shell 3.12.3
File Edit Shell Debug Options Window Help
Python 3.12.3 (tags/v3.12.3:f6650f9, Apr 9 2024, 14:05:25) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>> = RESTART: C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/RealEstate.py
1. Average Listing Price by Location:
Location
Bangalore      8500000.0
Chennai        7250000.0
Hyderabad      15000000.0
Name: Listing_Price, dtype: float64

2. Number of Properties with More Than Four Bedrooms:
2

3. Property with the Largest Area:
Property_ID      104
Location         Hyderabad
Bedrooms         6
Area_sqft        3000
Listing_Price    15000000
Name: 3, dtype: object
>>>
```

Experiment 10

1.Monthly Sales Visualization

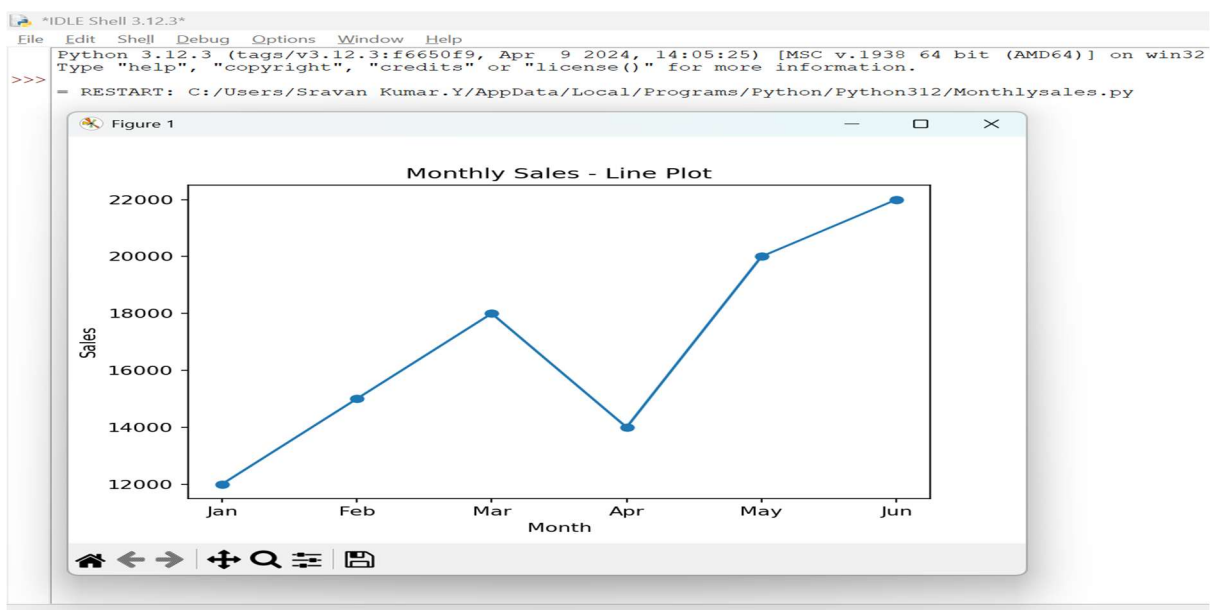
CODE

```
Monthlysales.py - C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/Monthlysales.py (3.12.3)
File Edit Format Run Options Window Help
import matplotlib.pyplot as plt

months = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun']
sales = [12000, 15000, 18000, 14000, 20000, 22000]

plt.plot(months, sales, marker='o')
plt.xlabel("Month")
plt.ylabel("Sales")
plt.title("Monthly Sales - Line Plot")
plt.show()
```

OUTPUT



2.Bar Plot of Monthly Sales

CODE

barplotofsales.py - C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/barplotofsales.py (3.12.3)

File Edit Format Run Options Window Help

```
import matplotlib.pyplot as plt

months = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun']
sales = [12000, 15000, 18000, 14000, 20000, 22000]

plt.bar(months, sales)
plt.xlabel("Month")
plt.ylabel("Sales")
plt.title("Monthly Sales - Bar Plot")
plt.show()
```

OUTPUT

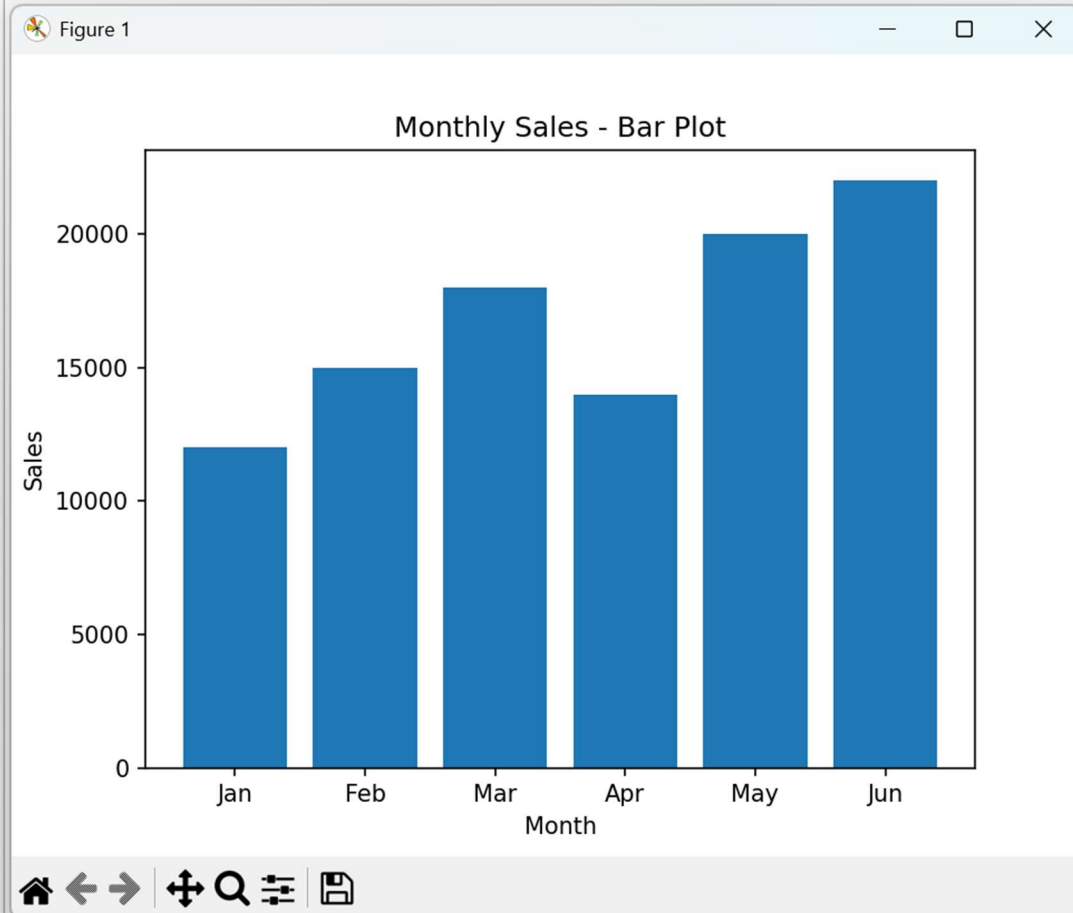
IDLE Shell 3.12.3

File Edit Shell Debug Options Window Help

Python 3.12.3 (tags/v3.12.3:f6650f9, Apr 9 2024, 14:05:25) [MSC v.1938 64 bit (AMD64)]
Type "help", "copyright", "credits" or "license()" for more information.

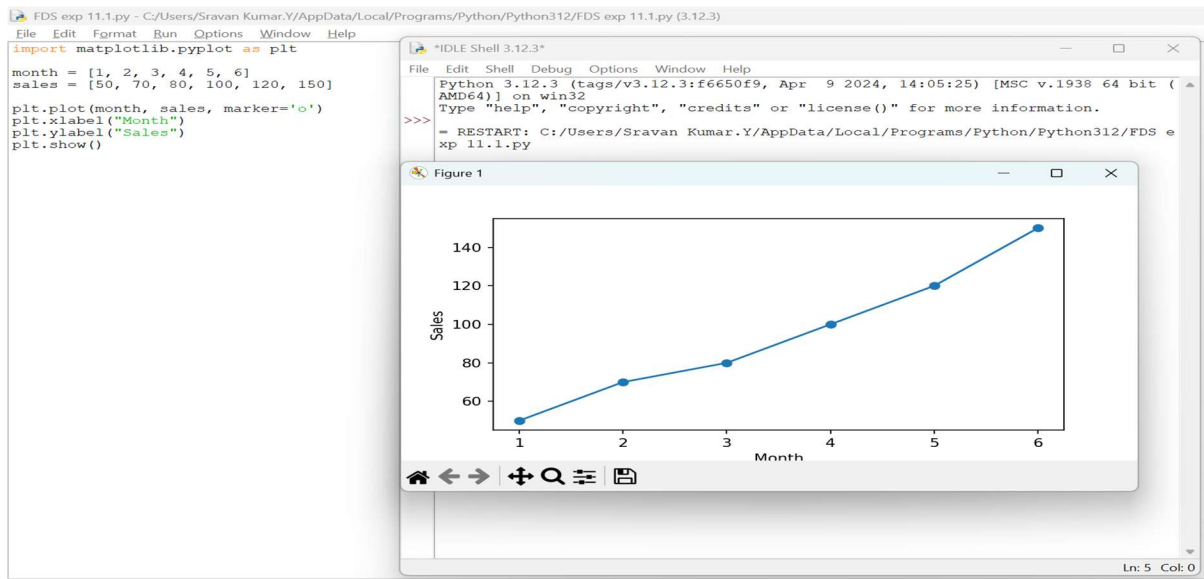
>>>

= RESTART: C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/barplotofs

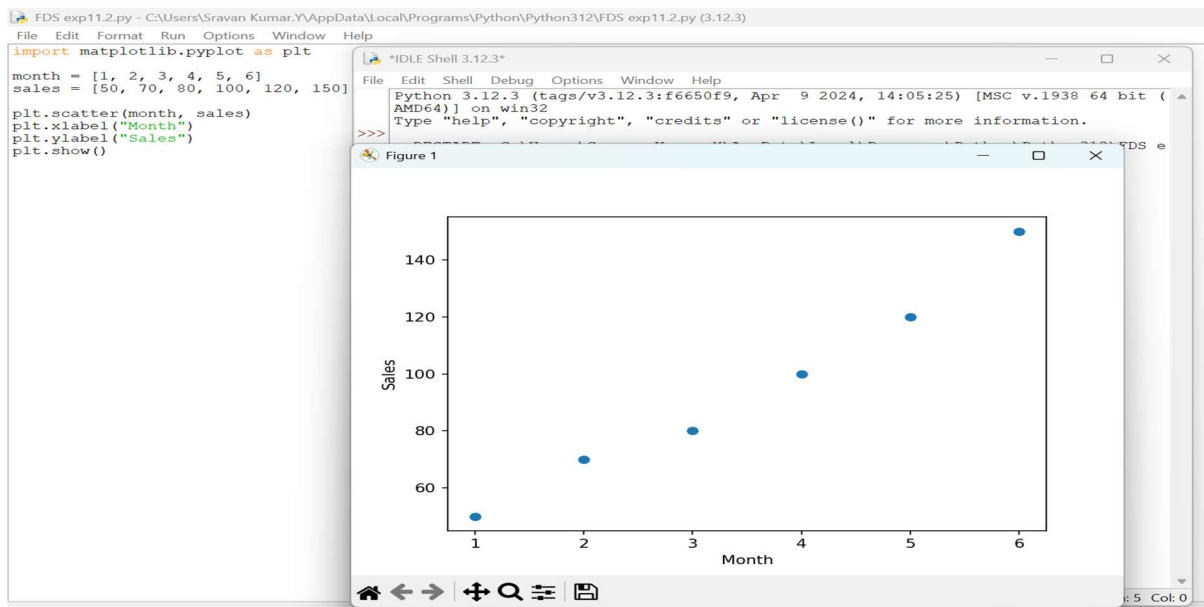


Experiment 11 Sales Visualization using Matplotlib

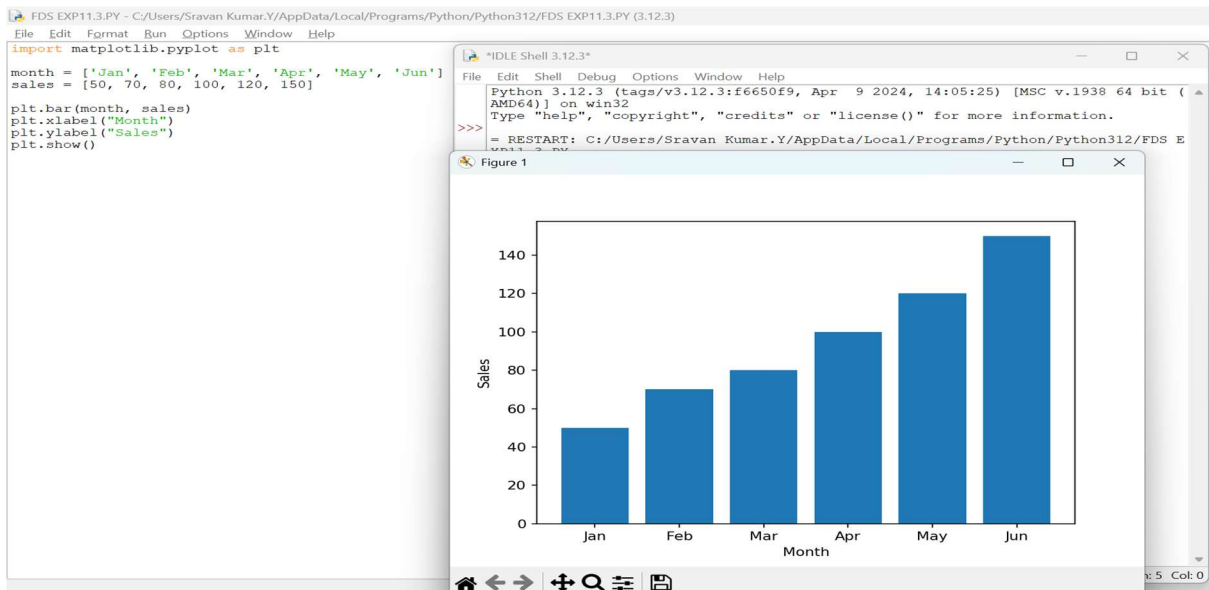
1. Simple Line Plot



2. Scatter Plot



3.Bar Plot



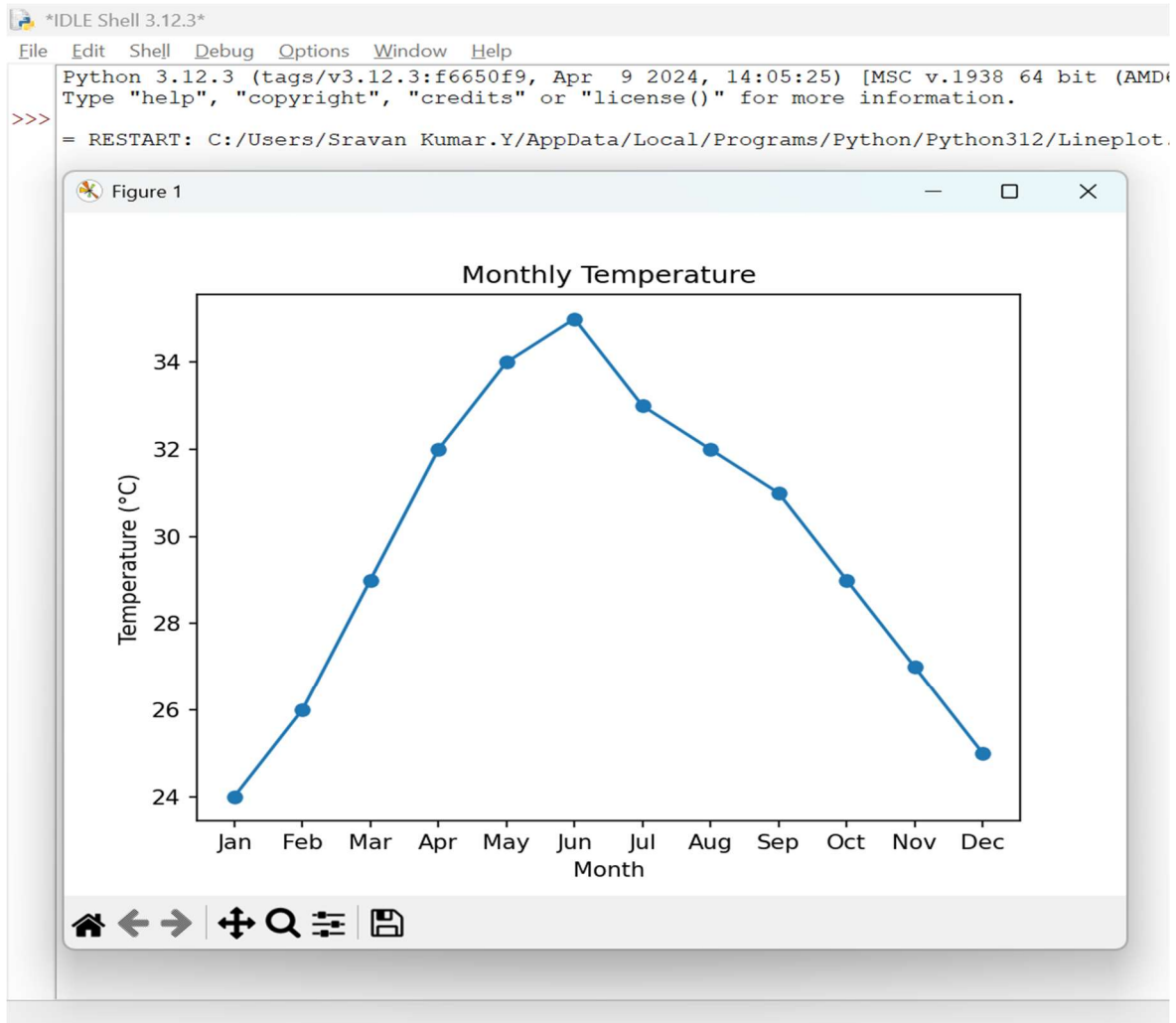
Experiment 12

1.Line Plot of Monthly Temperature

CODE

```
Lineplot.py - C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/Lineplot.py (3.12.3)
File Edit Format Run Options Window Help
import matplotlib.pyplot as plt
months = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun',
          'Jul', 'Aug', 'Sep', 'Oct', 'Nov', 'Dec']
temperature = [24, 26, 29, 32, 34, 35, 33, 32, 31, 29, 27, 25]
plt.plot(months, temperature, marker='o')
plt.xlabel("Month")
plt.ylabel("Temperature (°C)")
plt.title("Monthly Temperature")
plt.show()
```

OUTPUT



2. Scatter Plot of Monthly Rainfall

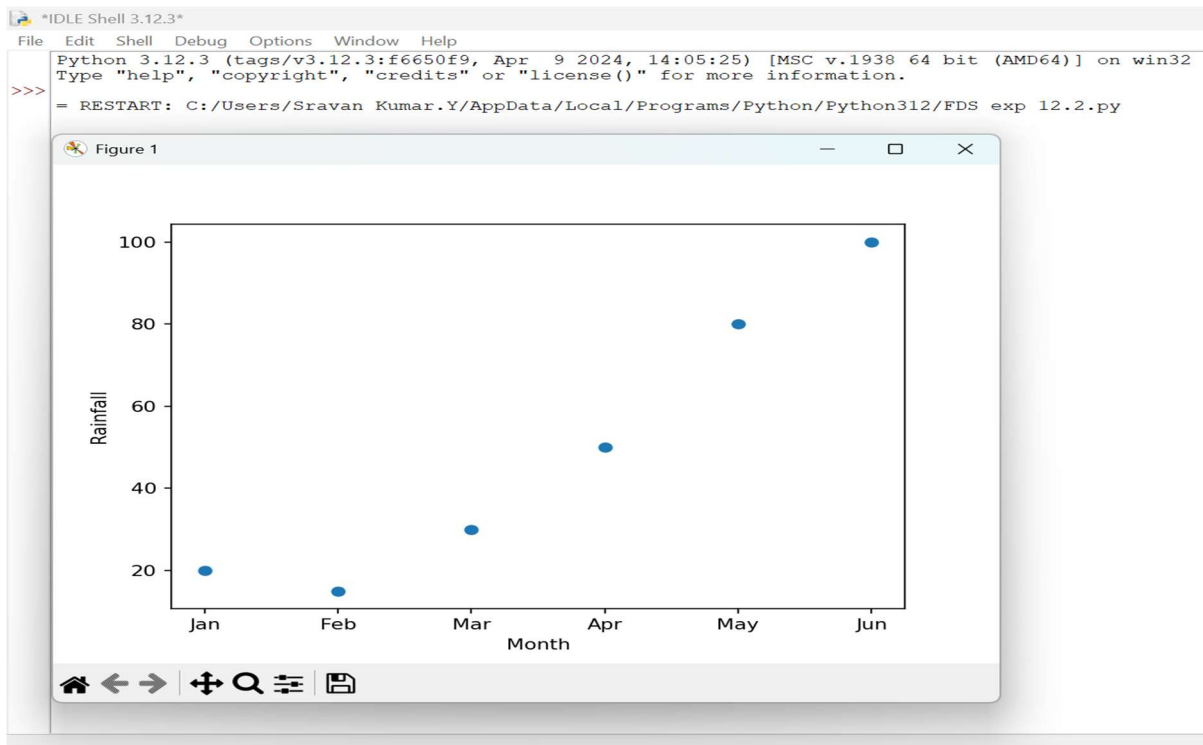
FDS exp 12.2.py - C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/FDS exp 12.2.py (3.12.3)

File Edit Format Run Options Window Help

```
import matplotlib.pyplot as plt

months = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun']
rainfall = [20, 15, 30, 50, 80, 100]

plt.scatter(months, rainfall)
plt.xlabel("Month")
plt.ylabel("Rainfall")
plt.show()
```



13. Word Frequency Distribution

CODE

WordfrequencyDis.py - C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/FDS Exp 13.py/WordfrequencyDis.py (3.12.3)

```
File Edit Format Run Options Window Help
from collections import Counter

file = open("sample_text.txt", "r")
text = file.read().lower()
file.close()

words = text.split()
frequency = Counter(words)

for word, count in frequency.items():
    print(word, ":", count)
```

IDLE Shell 3.12.3

File Edit Shell Debug Options Window Help

Python 3.12.3 (tags/v3.12.3:f6650f9, Apr 9 2024, 14:05:25) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.

>>> = RESTART: C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/FDS Exp 13.py/WordfrequencyDis.py

python : 2
is : 2
easy. : 1
powerful. : 1

>>> |

14. Frequency Distribution of Customer Ages

```
FDS Exp 14.py - C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/FDS Exp 14.py (3.12.3)
File Edit Format Run Options Window Help
import pandas as pd

data = {'Age': [20, 25, 20, 30, 25, 20, 35]}
df = pd.DataFrame(data)

print(df['Age'].value_counts().sort_index())
```

```
IDLE Shell 3.12.3
Python 3.12.3 (tags/v3.12.3:f6650f9, Apr 9 2024, 14:05:25) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/FDS Exp 14.py
Age
20    3
25    2
30    1
35    1
Name: count, dtype: int64
>>>
```

15. Frequency Distribution of Likes

```
FDS exp 15.py - C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/FDS exp 15.py (3.12.3)
File Edit Format Run Options Window Help
import pandas as pd

data = {'Likes': [10, 20, 10, 30, 20, 10, 40]}
df = pd.DataFrame(data)

print(df['Likes'].value_counts().sort_index())
```

```
IDLE Shell 3.12.3
Python 3.12.3 (tags/v3.12.3:f6650f9, Apr 9 2024, 14:05:25) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/FDS exp 15.py
Likes
10    3
20    2
30    1
40    1
Name: count, dtype: int64
>>>|
```